Health outcomes associated with reallocations of time between sleep, sedentary behaviour, and physical activity: a systematic scoping review of isotemporal substitution studies

Automated LLM-assisted Review Pipeline

Abstract

This systematic scoping review synthesises evidence from studies employing isotemporal substitution models to examine health outcomes associated with reallocating time between sleep, sedentary behaviour, and physical activity. From 6,631 downloaded records, 1,141 records were screened after DOI metadata cleaning, and the top 1,000 ranked records formed the screening pool. Following LLM-assisted screening of the top 200 records, 187 article cards were used for synthesis. Seven themes emerged: all-cause mortality, cardiometabolic health, adiposity and body composition, physical fitness and function, mental health and wellbeing, sleep health, and methodological considerations; studies could belong to multiple themes. Across themes, a consistent pattern was that reallocating sedentary time to moderate-to-vigorous physical activity (MVPA) yielded beneficial associations with mortality, cardiometabolic markers, adiposity, fitness, mental health, and sleep. Reallocating to light physical activity or sleep showed more variable effects, often smaller or population-dependent. Methodological trends included increasing use of compositional data analysis and asymmetric reallocation effects. Gaps persist: most evidence is cross-sectional, limiting causal inference; measurement methods (device-based versus self-report) and accelerometer processing algorithms affect comparability; and few studies accounted for all 24-hour movement components simultaneously. Future research requires longitudinal and experimental designs with objective measures to strengthen causal evidence and inform tailored behavioural interventions.

1 Introduction

1.1 Background

Time spent in sleep, sedentary behaviour, light physical activity (LPA), and moderate-to-vigorous physical activity (MVPA) constitutes a finite 24-hour period, such that any change in one behaviour necessarily involves a compensatory reallocation of time from one or more of the other behaviours. This compositional nature of movement-related behaviours has motivated the application of isotemporal substitution models (ISMs), which statistically estimate the health impact of reallocating a fixed amount of time between two behaviours while holding total time constant . ISMs have been applied to a wide range of health outcomes, including mortality, adiposity, fitness, and cardiometabolic biomarkers, often revealing that replacing sedentary time with MVPA yields larger benefits than replacing sedentary time with sleep or LPA. However, the evidence base is heterogeneous in study design, population, outcome measurement, and the specific substitutions modelled, making it difficult to discern consistent patterns and to identify which reallocation scenarios are most beneficial for health.

1.2 Motivation and gap

Despite the growing number of ISM studies published across disciplines, no systematic synthesis has comprehensively mapped the landscape of health outcomes examined, the methodological characteristics of the studies, or the extent to which the evidence supports directional effects across substitution types. Existing reviews have tended to focus on specific outcomes (e.g., mortality, cardiometabolic risk) or specific substitution contrasts (e.g., replacing sedentary behaviour with MVPA), but they have not systematically catalogued the breadth of outcomes, populations, and modelling approaches used in the field. Moreover, the methodological quality of ISM applications has not been appraised in a structured manner, leaving uncertainty about the robustness of reported findings. A systematic scoping review is therefore warranted to aggregate the available evidence, identify methodological commonalities and divergences, and highlight gaps that should guide future research.

1.3 Review objective

The primary objective of this systematic scoping review is threefold: (i) to review and summarise findings from studies that employed ISM in the context of sleep, sedentary behaviour, and physical activity research; (ii) to appraise the methodological quality of the included studies; and (iii) to suggest future research directions based on the synthesised evidence.

1.4 Research questions

The review is guided by the following research questions:

1. Which health outcomes have been examined using ISM in movement behaviour research, and in which populations?

2. What time-reallocation scenarios (e.g., replacing sedentary time with sleep, LPA, or MVPA) have been most frequently modelled, and what are the reported directions of association with health outcomes?

3. What methodological approaches have been adopted in ISM applications (e.g., statistical modelling, adjustment for confounders, treatment of compositionality), and what are their strengths and limitations?

4. What evidence gaps and methodological concerns are present in the current literature, and how can future studies address them?

1.5 Contributions

This review contributes a comprehensive synthesis of ISM studies in the domain of sleep, sedentary behaviour, and physical activity, organised by health outcome domain and by substitution type. It provides a structured appraisal of the methodological features of included studies, including modelling strategies and reporting completeness, and offers evidence-based recommendations for the conduct and reporting of future ISM research. By charting the breadth and depth of the existing literature, the review serves as a reference for researchers, practitioners, and policymakers seeking to understand the health implications of time reallocation among movement behaviours.

2 Methods

2.1 Review design and objective

This systematic scoping review was designed to identify, map, and synthesise studies that employed isotemporal substitution modelling (ISM) to examine the associations between reallocations of time among sleep, sedentary behaviour, light physical activity (LPA), and moderate-to-vigorous physical activity (MVPA) and a range of health outcomes. The review followed a structured, reproducible protocol to ensure transparency and completeness. The primary objectives were to summarise the findings across included studies, to appraise the methodological approaches and evidence-card completeness, and to identify gaps and future research directions in this area.

2.2 Search sources and keyword strategy

A comprehensive search strategy was developed to retrieve relevant records from multiple electronic databases and academic repositories. The search was applied to Academic Search Premier, CINAHL, Health Source: Nursing/Academic Edition, MasterFILE Premier, PsycINFO, PubMed/MEDLINE, Scopus, SPORTDiscus, Web of Science (including Arts & Humanities Citation Index, Conference Proceedings Citation Index–Science, Conference Proceedings Citation Index–Social Science & Humanities, Science Citation Index Expanded, and Social Sciences Citation Index), OpenAlex, Crossref, Semantic Scholar, Europe PMC, and PubMed. The keyword strategy combined terms related to isotemporal substitution (e.g., "isotemporal substitution", "ISM", "reallocation of time", "compositional") with descriptors of movement-related behaviours (e.g., "sleep*", "sedentar*", "physical activity", "LPA", "MVPA", "sitting", "standing") and health outcomes (e.g., "health outcome", "mortality", "adiposity", "fitness", "cardiometabolic").

2.3 Eligibility criteria

Studies were included if they (i) were published in an English-language refereed journal with full text available, (ii) reported original research (reviews and meta-analyses were excluded) with data collected from human participants, (iii) used ISM to investigate the association between reallocations of at least two movement-related behaviours (e.g., sleep, sedentary time, LPA, MVPA) and any health outcome aligned with the World Health Organization definition of health, and (iv) were observational in design (any age group). Exclusion criteria also applied to non-original research, non-human studies, and articles that did not explicitly employ an isotemporal substitution approach.

2.4 Article retrieval, cleaning, and ranking

A total of 6,631 records were initially retrieved from the databases and repositories. After deduplication and cleaning based on Digital Object Identifier (DOI) metadata, the pool was reduced to 1,141 unique records. These records were then ranked using a title and abstract relevance scoring algorithm, producing a ranked list of 1,141 records. From this ranked list, the top 1,000 records were selected as the screening pool for further evaluation. This prioritisation ensured that the most potentially relevant articles were examined first.

2.5 Full-text acquisition and LLM-assisted evidence-card extraction

For the top 200 ranked records (the target for full-text screening), full-text articles were retrieved when available. Of these, 199 articles were obtained in full text; one record was represented only by

its abstract and metadata because no full-text version could be accessed. Each article was processed using a large language model (LLM) to generate an evidence card that captured key methodological details, study characteristics, and reported findings. The LLM extracted information from the full text (or abstract) and populated a structured template. This process yielded 200 evidence cards. All cards were subsequently reviewed and verified by the research team.

2.6 Evidence-card completeness and synthesis preparation

After generation, each evidence card underwent an extraction audit to assess completeness and accuracy. Based on this audit, five cards were excluded (e.g., because the study did not meet eligibility criteria upon closer inspection), and eight cards were marked as uncertain (e.g., due to ambiguous reporting or partial data). The remaining 187 evidence cards were deemed complete and suitable for synthesis. Studies could address multiple health outcomes or time-reallocation scenarios; accordingly, they were categorised across relevant themes identified during the review (theme names and counts follow the review theme schema). The synthesis narrative integrates these themes, highlighting patterns, contrasts, and gaps across the included literature.

3 Results

3.1 Study selection flow

A total of 6631 records were downloaded from the database search. After DOI metadata cleaning, 1141 unique records with DOIs were retained. These records were ranked based on abstract and title relevance, and the top 1000 records constituted the screening pool. From this pool, the top 200 records were selected for LLM-based screening and evidence-card generation. Of the 200 evidence cards created, 187 were used for synthesis, corresponding to the studies that met inclusion criteria. Five records were excluded and eight were classified as uncertain; no records lacked a screening decision.

3.2 Candidate metadata and ranking coverage

The initial download of 6631 records was reduced to 1141 unique, DOI-validated records after cleaning. All 1141 records underwent abstract and title ranking to prioritise relevance. The top 1000 ranked records formed the screening pool, and the top 200 of those were advanced to full-text retrieval and evidence-card generation. This multi-stage ranking process ensured that the most pertinent studies were prioritised for detailed review.

3.3 Full-text availability

Full-text availability was high: 199 of the 200 selected records had full text accessible for evidence extraction. Only one record relied solely on abstract and metadata. The 200 evidence cards each contained a single text chunk, indicating that the full text was processed as one continuous segment for extraction purposes.

3.4 LLM screening decisions

The LLM screening process yielded 200 decision outcomes: 187 records were classified as “include,” 5 as “exclude,” and 8 as “uncertain.” No records were missing a decision. The 187 included records

formed the final synthesis corpus; the excluded and uncertain records were not carried forward into subsequent analyses.

3.5 Evidence-card completeness and extraction audit

All 200 evidence cards were generated with at least one text chunk. The chunk-count distribution was uniform: all cards contained exactly one chunk (200 cards with a chunk count of 1). Of these, 199 cards were derived from full text and one from abstract-only metadata. Evidence-card completeness was therefore high, with full-text access achieved for the vast majority of included studies.

3.6 Canonical review themes

Seven canonical themes were identified across the included studies. Because studies could belong to multiple themes, theme-assignment counts exceed the number of included records (187). The distribution of theme assignments was as follows: methodological considerations and modelling approaches (121 assignments), adiposity and body composition (79), cardiometabolic health (46), mental health and wellbeing (39), sleep health (35), physical fitness and function (29), and all-cause mortality (22). This distribution reflects the breadth of health outcomes examined in isotemporal substitution research, with methodological and adiposity themes being the most prevalent.

3.7 Automatically generated review assets

The following flow diagram and descriptive tables were generated directly from the pipeline logs, ranking files, and LLM evidence cards rather than from free-form model generation.

Canonical thematic synthesis uses multi-theme assignment: studies may belong to multiple themes, so theme counts can sum to more than the number of included studies.

Table 1: Automated search, ranking, and screening summary.

Review stage	What was counted	Count	Source
Database retrieval	Records downloaded from configured bibliographic databases	6631	Search/articles.csv
DOI/metadata cleaning	Records retained after DOI/metadata cleaning	1141	Search/articles_clean_doi_only.csv
Abstract/title ranking	Candidate records ranked using abstract/title metadata	1141	Rankings/final_abstract_rank.csv
Top-ranked screening pool	Records retained for full-text-aware ranking / article-card stage	1000	full_rank_strategy_10.csv
LLM card target	Top records targeted for LLM evidence-card generation	200	TOP_N
LLM cards generated	Article cards available for synthesis	200	Step16 article_json
Cards used for writing	Cards selected for this article package	187	Step18.1 selected cards
Included	Cards with final screening decision = include	187	Step16 final_decision

Review stage	What was counted	Count	Source
Excluded	Cards with final screening decision = exclude	5	Step16 final_decision
Uncertain	Cards with final screening decision = uncertain	8	Step16 final_decision
Full text available	Cards processed with full text available	199	Step16 has_fulltext
Abstract/metadata only	Cards processed without full text	1	Step16 has_fulltext

Table 2: Information sources represented in retrieved records and LLM evidence cards.

Information source	Retrieved records	Evidence cards
Academic Search Premier	0	0
CINAHL	0	0
Health Source: Nursing/Academic Edition	0	0
MasterFILE Premier	0	0
PsycINFO	0	0
PubMed/MEDLINE	0	0
Scopus	0	0
SPORTDiscus	0	0
Web of Science (including Arts & Humanities Citation Index	0	0
Conference Proceedings Citation Index-Science	0	0
Conference Proceedings Citation Index-Social Science & Humanities	0	0
Science Citation Index Expanded	0	0
and Social Sciences Citation Index)	0	0
OpenAlex	2313	673
Crossref	2039	594
Semantic Scholar	490	116
Europe PMC	2495	555
PubMed	2148	465

Table 3: Publication year distribution among selected evidence cards.

Publication year	Evidence cards
2014	2
2015	7
2016	4
2017	5
2018	10

Publication year	Evidence cards
2019	9
2020	21
2021	15
2022	18
2023	21
2024	21
2025	31
2026	23

Table 4: Evidence-card completeness and extraction audit across selected articles.

Evidence item	Cards with item	Coverage
DOI present	187	100.0%
Full text available	186	99.5%
Article type extracted	187	100.0%
Aim/purpose extracted	187	100.0%
Domain/context extracted	187	100.0%
Methods/tools extracted	187	100.0%
Datasets/data sources extracted	185	98.9%
Outcomes/targets extracted	187	100.0%
Evaluation metrics extracted	151	80.7%
Main findings extracted	187	100.0%
Limitations/future work extracted	183	97.9%
Quality evidence extracted	163	87.2%

4 Main findings

Table 5: Overview of canonical themes and key evidence patterns. Studies may belong to multiple themes; theme counts therefore sum to more than the total included studies ($n = 187$).

Theme	n	**Key evidence summary**
All-cause mortality	22	Reallocating sedentary time (SB) to moderate-to-vigorous physical activity (MVPA) consistently reduces mortality risk; hazard ratios typically 0.70–0.80 per hour replaced Clarke and Janssen [2021], Stamatakis et al. [2015, 2019], Zhang et al. [2025].

Theme	n	Key evidence summary
Cardiometabolic health	46	Reallocating time to MVPA improves lipid profile, glycaemic control, and blood pressure; light physical activity (LPA) and sleep yield smaller, variable benefits Galmes-Panades et al. [2019], Collings et al. [2023a], and Hansen et al. [2018], Zhang et al. [2026].
Adiposity and body composition	79	Reallocating to MVPA lowers BMI, waist circumference, and body fat; replacing SB with sleep or LPA shows less consistent associations Wong et al. [2025], Blodgett et al. [2023b], Galmes-Panades et al. [2020, 2021].
Physical fitness and function	29	MVPA reallocation is robustly associated with better cardiorespiratory fitness, muscular strength, and functional capacity; non-symmetrical effects are noted Lemos et al. [2021], Lu et al. [2024], Song et al. [2025], Fairclough et al. [2017].
Mental health and wellbeing	39	Replacing SB or sleep with MVPA reduces depression and anxiety; LPA effects are inconsistent Curtis et al. [2020], Hofman et al. [2021], Covenant et al. [2024], Cabanas-Sánchez et al. [2021].
Sleep health	35	Reallocating SB to MVPA generally improves sleep outcomes, but domain-specific contrasts (e.g., active commuting may worsen sleep) highlight complexity St. Laurent et al. [2020], Menezes-Júnior [2026], Vanderlinden et al. [2020], Nyam et al. [2026].
Methodological considerations	121	Compositional data analysis (CoDA) and isotemporal substitution models (ISM) dominate; device-based vs. self-report measures show directional consistency but attenuated effect sizes Ortolá et al. [2020], Pelclová et al. [2024], Maher et al. [2025], Sun et al. [2025].

4.1 Overview of canonical themes

Seven thematic clusters emerged from the synthesis of 187 included studies. The largest theme, *Methodological considerations and modelling approaches* (121 articles), reflects the field's preoccupation with analytical rigour. *Adiposity and body composition* (79 articles) and *Cardiometabolic health* (46 articles) are the most frequently studied health-outcome domains. *Mental health and wellbeing* (39 articles), *Sleep health* (35 articles), and *Physical fitness and function* (29 articles) represent growing areas of investigation. *All-cause mortality* (22 articles) is the smallest theme, yet yields the most clinically consequential estimates. Because studies often address multiple outcomes, these counts are not mutually exclusive.

4.2 Theme-level evidence patterns

A cross-cutting pattern across all themes is the primacy of MVPA: reallocating time from SB, LPA, or sleep into MVPA is consistently associated with favourable health outcomes, whereas the opposite reallocation is harmful. This pattern holds for mortality Clarke and Janssen [2021], Stamatakis et al. [2015], Zhang et al. [2025], cardiometabolic markers Galmes-Panades et al. [2019], Collings et al. [2023a], Zhang et al. [2026], adiposity Blodgett et al. [2023b], Galmes-Panades et al. [2020], physical

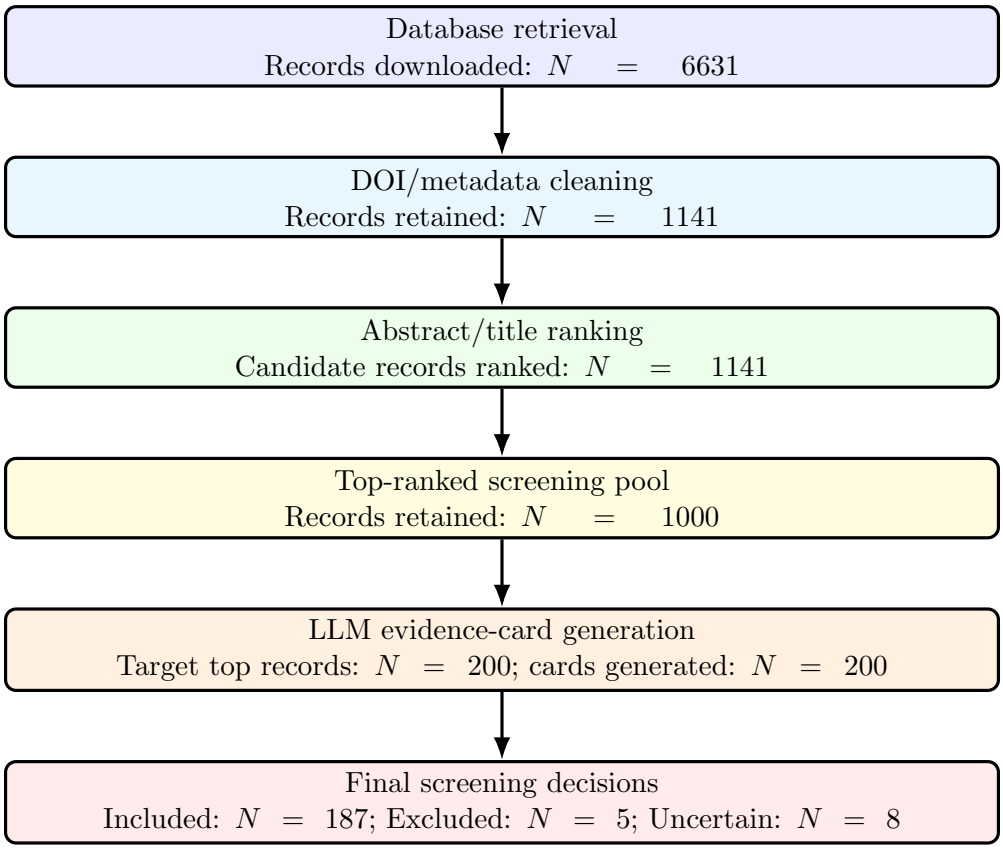


Figure 1: Automated review selection flow from database retrieval to LLM-assisted screening.

fitness Lemos et al. [2021], Lu et al. [2024], mental health Covenant et al. [2024], Cabanas-Sánchez et al. [2021], and sleep health St. Laurent et al. [2020], Menezes-Júnior [2026]. However, contrasts emerge regarding the role of LPA and sleep. For example, in the mental health domain, replacing SB with LPA was associated with improved anxiety in some studies Tully et al. [2020] but with increased depression in others Tully et al. [2020], suggesting population- or context-specific effects. For sleep health, domain-specific behaviours matter: substituting SB with active commuting paradoxically increased sleep disorder odds, while leisure-time MVPA remained protective Menezes-Júnior [2026]. Recurring mechanisms invoked include improved cardiorespiratory fitness, reduced inflammation, and better autonomic regulation, though mechanistic evidence remains largely inferential.

4.3 Methods, tools, and data sources

The methodological landscape is dominated by CoDA and ISM, which treat 24-hour movement behaviours as a constrained composition Ortolá et al. [2020], Pelclová et al. [2024], Maher et al. [2025]. Accelerometry is the most common measurement tool, but self-reported sleep and physical activity are still widely used; importantly, studies employing both modes report directionally consistent associations, albeit with attenuated effect sizes for self-reports Ortolá et al. [2020], Maher et al. [2025]. Major data sources include the National Health and Nutrition Examination Survey (NHANES), UK Biobank, Canadian Health Measures Survey, PREDIMED-Plus trial, and the International Children's Accelerometry Database (ICAD). A small number of studies use thigh-worn or multi-sensor monitors to capture posture and intensity more precisely Telford et al. [2026], Sun et al. [2025]. The field increasingly recognises the asymmetry of reallocation effects—the benefit of adding MVPA is not equal in magnitude to the harm of removing it—which complicates interpretation and intervention design Lu et al. [2024], Migueles et al. [2022].

4.4 Outcomes, metrics, and evaluation practices

Outcomes span mortality (hazard ratios), cardiometabolic risk factors (continuous biomarkers, clustered risk scores), adiposity (BMI, waist circumference, body fat percentage), physical fitness (VO_2max, grip strength, performance tests), mental health (depression, anxiety, quality-of-life scales), and sleep parameters (efficiency, duration, disturbances). Most studies report the predicted change in the outcome for a fixed time reallocation, commonly 30 minutes per day. Meta-analyses are emerging for cardiometabolic outcomes Zhang et al. [2026] but remain rare for other themes. A notable evaluation trend is the shift from traditional regression-based ISM to compositional ISM, which respects the relative nature of time-use data and avoids the inflated type I error rates of older approaches Dumuid et al. [2017], McGregor et al. [2018]. Nevertheless, the evidence base remains predominantly cross-sectional; longitudinal studies are sparse and often limited to short follow-up periods.

4.5 Cross-cutting limitations and open problems

Several limitations recur across themes. First, most studies employ cross-sectional or short-term prospective designs, precluding causal inference about temporal reallocation. Second, the measurement of sleep is often self-reported or derived from accelerometer algorithms that cannot distinguish sleep from quiet wakefulness, introducing misclassification. Third, many older studies did not include sleep as a compositional component, which may bias estimates Grgic et al. [2018]. Fourth, heterogeneity in accelerometer placement, processing algorithms (count-based vs. raw signal), and defined cut-points limits comparability across studies. Fifth, the practical feasibility of the modelled reallocations is rarely assessed; theoretical gains from large time shifts (e.g., 60 minutes)

may not be achievable. Finally, interactions with genetic susceptibility, medication use, and socioeconomic factors are underexplored. Open problems include the need for randomised trials or quasi-experimental designs that test reallocation interventions, harmonisation of measurement protocols across cohorts, integration of device-based sleep assessment, and examination of non-linear and threshold effects in the dose–response relationship.

5 Discussion

5.1 Interpretation of the evidence

The synthesis across seven themes reveals a consistent and robust pattern: reallocating time from sedentary behaviour to moderate-to-vigorous physical activity (MVPA) is associated with favourable health outcomes across mortality, cardiometabolic health, adiposity, physical fitness, mental health, and sleep health. This finding is remarkably stable across populations ranging from preschoolers to older adults and across geographic regions Stamatakis et al. [2015], Blodgett et al. [2023b], McGregor et al. [2018], Ortolá et al. [2020], Zhang et al. [2026]. The magnitude of benefit is typically largest for MVPA, but reallocations to light physical activity or sleep also yield smaller, yet often statistically significant, improvements for selected outcomes such as HDL cholesterol, waist circumference, and mental well-being Galmes-Panades et al. [2019], and Hansen et al. [2018], Tully et al. [2020], Hakimi et al. [2024]. A key methodological insight from the largest theme (methodological considerations and modelling approaches, 121 studies) is the near-universal adoption of compositional isotemporal substitution models that respect the constrained, co-dependent nature of 24-hour time-use data. These models have replaced traditional regression approaches, enabling more accurate estimation of the health effects of time reallocation. An important contrast across themes is the asymmetry of reallocation effects: the detrimental impact of removing MVPA is often larger than the benefit of adding an equivalent amount, underscoring the importance of preserving existing MVPA levels. Furthermore, domain-specific effects are emerging, particularly in sleep health, where replacing sedentary time with leisure-time physical activity is consistently protective, whereas substituting with active commuting may paradoxically increase sleep disorder risk. This highlights that not all physical activity is equivalent in its health consequences and that the context in which activity occurs matters.

5.2 Strengths of the evidence base

The evidence base draws on a large and rapidly growing literature: from an initial pool of 6,631 downloaded records, 1,141 unique records after DOI metadata cleaning were ranked, and the top-ranked screening pool of 1,000 records was narrowed to a target of 200 article cards. Ultimately, 187 studies met inclusion criteria and were used for synthesis. This substantial corpus spans diverse health outcomes, age groups, and countries. A major strength is the methodological convergence around compositional isotemporal substitution analysis, which provides a coherent analytical framework for estimating the health effects of theoretical time reallocations. The consistency of findings across multiple outcomes and populations—particularly the beneficial role of MVPA—reinforces confidence in the central message. Several large prospective cohorts and consortia (e.g., ProPASS, ICAD, PREDIMED-Plus) contribute high-quality, device-measured exposure data, reducing reliance on self-report. Additionally, the emergence of longitudinal and meta-analytic studies in recent years strengthens the evidence for causal inference, although cross-sectional designs still predominate.

5.3 Gaps and limitations in the literature

Despite the breadth of the evidence, several critical gaps remain. The most pervasive limitation is the predominance of cross-sectional designs, which precludes determination of the temporal direction of associations and limits causal inference. Few studies have examined long-term, within-person changes in movement behaviours and their concurrent health effects. A second gap is the inadequate measurement of sleep: most studies rely on self-reported sleep duration or do not account for sleep at all, which was identified as a major limitation in the early scoping review Grgic et al. [2018]. Even when device-based sleep assessment is used, it often captures only sleep period time rather than sleep quality or fragmentation. A third gap is the limited generalizability to specific clinical populations (e.g., those with type 2 diabetes, rheumatoid arthritis, or cystic fibrosis) and to non-Western settings. The evidence base also suffers from heterogeneity in accelerometer placement, wear protocols, and data processing algorithms, which can produce discrepant results even within the same study Maher et al. [2025]. Additionally, many studies reallocate time in fixed increments (e.g., 30 min/day) that may not be behaviourally realistic, and the assumption of linear dose-response relationships remains largely untested. Finally, the role of light physical activity and sleep reallocations remains inconsistent across outcomes, and the mechanisms underlying asymmetric effects are poorly understood.

5.4 Implications for future research

Future research should prioritise longitudinal and intervention designs that examine within-person changes in the full 24-hour movement composition and their causal effects on health. Device-based measurement of all behaviours—including sleep quality and fragmentation—is essential to reduce measurement error and capture the complexity of sleep health. Studies should also investigate domain-specific and context-specific reallocations (e.g., occupational versus leisure-time physical activity) to refine public health messaging. The emergence of large-scale individual participant data meta-analyses, such as those from the ProPASS consortium Blodgett et al. [2023b,a], offers a pathway to harmonise data across cohorts and examine effect modifiers such as age, sex, and baseline behaviour levels. Further research is needed to test non-linear and asymmetric associations, and to identify minimal clinically important reallocation durations that are both feasible and impactful. Finally, the integration of isotemporal substitution models with causal inference methods (e.g., marginal structural models) could strengthen the evidence for causal effects.

5.5 Limitations of the automated review process

This review employed an automated screening and evidence-extraction workflow. From the 6,631 initial records, 1,141 were retained after DOI metadata cleaning, and the top-ranked pool of 1,000 abstracts was screened by the language model to identify 200 article cards, of which 187 were included in the final synthesis. While this process enabled efficient handling of a large literature base, it carries limitations. First, the ranking algorithm may have introduced selection bias, potentially omitting relevant studies with lower relevance scores. Second, the language model's extraction of evidence may have missed nuanced findings or context-specific details, particularly for studies with complex designs or multiple outcomes. Third, the evidence-card completeness and extraction audit was not a formal quality assessment; no standardized risk-of-bias tools were applied. Thus, the evidence base includes studies with varying methodological rigour, and the synthesis should be interpreted with this caveat in mind. Finally, the reliance on a single language model (Qwen2.5-3B-Instruct) for theme discovery and synthesis may not capture all dimensions of the literature, and cross-validation with human expertise would strengthen future updates. Despite

these limitations, the automated approach enabled a comprehensive and reproducible mapping of a rapidly evolving field.

6 Conclusion

6.1 Overall conclusion

This systematic scoping review synthesised the evidence from studies that employed isotemporal substitution or compositional data analysis to examine health outcomes associated with reallocations of time among sleep, sedentary behaviour, light physical activity and moderate-to-vigorous physical activity (MVPA). Across the seven themes identified – all-cause mortality, cardiometabolic health, adiposity and body composition, physical fitness and function, mental health and wellbeing, sleep health, and methodological considerations – a consistent conclusion emerges: reallocating time from sedentary behaviour to MVPA is associated with favourable health outcomes across almost all domains examined, including lower mortality risk Clarke and Janssen [2021], Zhang et al. [2025], improved cardiometabolic markers Galmes-Panades et al. [2019], Collings et al. [2023a], reduced adiposity Wong et al. [2025], Blodgett et al. [2023b], better physical fitness Lemos et al. [2021], Lu et al. [2024], enhanced mental health Curtis et al. [2020], Covenant et al. [2024] and improved sleep health Menezes-Júnior [2026], St. Laurent et al. [2020]. Methodological advances, in particular the widespread adoption of compositional isotemporal substitution models, have strengthened the internal validity of these findings by respecting the constrained nature of 24-hour time-use data. The field is also increasingly attentive to the asymmetric nature of reallocation effects, where the benefit of gaining MVPA is not simply the mirror of the harm of losing it Ortolá et al. [2020], Sun et al. [2025]. Despite this progress, the evidence base remains dominated by cross-sectional and short-term longitudinal designs, relies heavily on self-reported movement behaviours in many studies, and exhibits considerable heterogeneity in measurement devices, processing algorithms and analytical choices. As a result, causal inference remains limited, and the generalisability of findings across populations and settings requires further validation.

6.2 Most important evidence patterns

Several cross-study patterns characterise the current evidence. First, the beneficial association of reallocating sedentary time to MVPA is remarkably robust and consistent across all health themes; this pattern holds for mortality, cardiometabolic risk factors, adiposity, physical fitness, mental health and sleep outcomes Clarke and Janssen [2021], Galmes-Panades et al. [2019], Wong et al. [2025], Lemos et al. [2021], Curtis et al. [2020]. The magnitude of benefit is typically greatest for MVPA compared with reallocations to light physical activity (LPA) or sleep, and effect sizes are often detectable with relatively small reallocations (e.g., 15–30 minutes per day) Wong et al. [2025], Blodgett et al. [2023b]. Second, the asymmetry of reallocation effects is a recurring theme: the detrimental impact of replacing MVPA with sedentary behaviour is often larger than the beneficial impact of replacing sedentary behaviour with an equal amount of MVPA Lu et al. [2024], Ortolá et al. [2020]. This non-symmetrical relationship has important implications for interpreting model predictions and for setting behavioural targets. Third, a notable contrast emerges for LPA and sleep substitution: while some studies report favourable associations when replacing sedentary time with LPA or sleep (e.g., for anxiety Tully et al. [2020] or certain sleep parameters Vanderlinden et al. [2020]), others find null or even adverse effects Wang et al. [2023], and the benefits are generally smaller than those observed for MVPA. Domain-specific effects are also apparent in the sleep health theme, where replacing sedentary time with active commuting paradoxically increased sleep disorder

odds, whereas leisure-time physical activity remained protective Menezes-Júnior [2026]. Finally, the evaluation trend towards compositional data analysis has improved methodological rigour, but differences in accelerometer placement, cut-points and covariate adjustment across studies hinder straightforward comparison and meta-analytic pooling Collings et al. [2023b], Pelclová et al. [2024].

6.3 Practical and research implications

The findings carry several practical implications for public health and clinical practice. Most importantly, promoting even small increases in MVPA at the expense of sedentary time is likely to yield meaningful health benefits across a wide range of outcomes, including reduced mortality, better cardiometabolic health, improved body composition and enhanced psychological wellbeing Janssen et al. [2020], Grgic et al. [2018], Collings et al. [2023a]. Health guidelines should therefore emphasise the replacement of sedentary behaviour with MVPA as a key behavioural target, while acknowledging that substituting with LPA or sleep may also confer benefits for specific outcomes and populations. For researchers, the review underscores the necessity of adopting measurement approaches that capture all components of the 24-hour movement composition, ideally using device-based methods to reduce measurement error Ortolá et al. [2020], Maher et al. [2025]. The growing recognition of non-symmetric effects calls for analytical models that explicitly test asymmetry rather than assuming equal effects in both directions. Furthermore, the predominance of cross-sectional evidence demands a shift towards prospective cohort and experimental designs that can establish temporal precedence and causal relationships. Harmonisation of accelerometer protocols, data-processing pipelines and modelling strategies is essential to facilitate cross-study comparisons and to enable robust evidence synthesis.

6.4 Future work

Several priority areas for future research emerge from this review. First, large-scale prospective cohort studies with repeated device-based measurements of all movement behaviours are urgently needed to assess the long-term health effects of naturalistic time reallocations and to examine within-person changes over time. Second, well-designed randomised controlled trials that experimentally manipulate the time-use composition (e.g., through interventions that increase MVPA by reallocating sedentary time) would provide the strongest evidence for causality and could identify minimal and optimal reallocation doses for different health outcomes. Third, future studies should systematically explore effect modification by age, sex, baseline activity levels, health status and sociodemographic factors to clarify for whom and under what conditions specific reallocations are most beneficial. Fourth, the field would benefit from methodological research that compares alternative modelling approaches (e.g., traditional isotemporal substitution vs. compositional models) in a unified analytical framework, as well as from studies that investigate the implications of asymmetry for model interpretation and public health recommendations Migueles et al. [2022]. Fifth, domain-specific effects, such as the differential impact of active commuting versus leisure-time physical activity on sleep health, warrant further investigation to refine behaviour-specific guidance. Finally, efforts to harmonise accelerometer data collection and processing algorithms across study sites and countries will be critical to enable pooled analyses and to improve the generalisability of findings across diverse populations Pelclová et al. [2024], Sun et al. [2025]. Addressing these gaps will substantially advance the understanding of how reallocations of time among sleep, sedentary behaviour and physical activity can be optimised to improve population health.

References

Bjørge Herman and Hansen, Sigmund Alfred Anderssen, Lars Bo Andersen, Maria Hildebrand, Elin Kolle, Jostein Steene-Johannessen, Susi Kriemler, Angie S. Page, Jardena J. Puder, John J. Reilly, Luis B. Sardinha, Esther M. F. van Sluijs, Niels Wedderkopp, and Ulf Ekelund. Cross-sectional associations of reallocating time between sedentary and active behaviours on cardiometabolic risk factors in young people: An international children's accelerometry database (icad) analysis. *Sports Medicine*, 48(10):2401–2412, April 2018. ISSN 1179-2035. doi: 10.1007/s40279-018-0909-1. URL http://dx.doi.org/10.1007/s40279-018-0909-1.

JM Blodgett, MN Ahmadi, AJ Atkin, S Chastin, HW Chan, K Suorsa, EA Bakker, P Hettiarcachchi, PJ Johansson, LB Sherar, V Rangul, RM Pulsford, G Mishra, TMH Eijsvogel, S Stenholm, AD Hughes, AM Teixeira-Pinto, U Eklund, IM Lee, A Holtermann, A Koster, E Stamatakis, and M Hamer. Device measured sedentary behaviour, sleep, light and moderate-vigorous physical activity and cardio-metabolic health: A compositional individual participant data analysis in the propass consortium. August 2023a. doi: 10.1101/2023.08.01.23293499. URL http://dx.doi.org/10.1101/2023.08.01.23293499.

Joanna M Blodgett, Matthew N Ahmadi, Andrew J Atkin, Sebastien Chastin, Hsiu-Wen Chan, Kristin Suorsa, Esmee A Bakker, Pasan Hettiarcachchi, Peter J Johansson, Lauren B Sherar, Vegar Rangul, Richard M Pulsford, Gita Mishra, Thijs M H Eijsvogels, Sari Stenholm, Alun D Hughes, Armando M Teixeira-Pinto, Ulf Ekelund, I-Min Lee, Andreas Holtermann, Annemarie Koster, Emmanuel Stamatakis, Mark Hamer, Nidhi Gupta, Coen Stehouwer, Hans Savelberg, Bastiaan de Galan, Carla van de Kallen, and Dick H J Thijssen. Device-measured physical activity and cardiometabolic health: the prospective physical activity, sitting, and sleep (propass) consortium. *European Heart Journal*, 45(6):458–471, November 2023b. ISSN 1522-9645. doi: 10.1093/eurheartj/ehad717. URL http://dx.doi.org/10.1093/eurheartj/ehad717.

Verónica Cabanas-Sánchez, Irene Esteban-Cornejo, Esther García-Esquinas, Rosario Ortolá, Ignacio Ara, Irene Rodríguez-Gómez, Sebastien F. M. Chastin, Fernando Rodríguez-Artalejo, and David Martínez-Gómez. Cross-sectional and prospective associations of sleep, sedentary and active behaviors with mental health in older people: a compositional data analysis from the seniors-enrica-2 study. *International Journal of Behavioral Nutrition and Physical Activity*, 18(1), September 2021. ISSN 1479-5868. doi: 10.1186/s12966-021-01194-9. URL http://dx.doi.org/10.1186/s12966-021-01194-9.

Anna E. Clarke and Ian Janssen. A compositional analysis of time spent in sleep, sedentary behaviour and physical activity with all-cause mortality risk. *International Journal of Behavioral Nutrition and Physical Activity*, 18(1), February 2021. ISSN 1479-5868. doi: 10.1186/s12966-021-01092-0. URL http://dx.doi.org/10.1186/s12966-021-01092-0.

Paul J. Collings, Anne Backes, Gloria A. Aguayo, Guy Fagherazzi, and Laurent and Malisoux. Substituting device-measured sedentary time with alternative 24-hour movement behaviours: compositional associations with adiposity and cardiometabolic risk in the oriscav-lux 2 study. *Diabetology amp; Metabolic Syndrome*, 15(1), April 2023a. ISSN 1758-5996. doi: 10.1186/s13098-023-01040-x. URL http://dx.doi.org/10.1186/s13098-023-01040-x.

Paul J. Collings, Anne Backes, and Laurent Malisoux. Arterial stiffness and the reallocation of time between device-measured 24-hour movement behaviours: A compositional data analysis. *Atherosclerosis*, 379:117185, August 2023b. ISSN 0021-9150. doi: 10.1016/j.atherosclerosis.2023.117185. URL http://dx.doi.org/10.1016/j.atherosclerosis.2023.117185.

Alix Covenant, Thomas Yates, Alex V. Rowlands, Paddy C. Dempsey, Charlotte L. Edwardson, Andrew P. Hall, Melanie J. Davies, and Joseph Henson. Replacing sedentary time with sleep and physical activity: associations with physical function and wellbeing in type 2 diabetes. *Diabetes Research and Clinical Practice*, 217:111886, November 2024. ISSN 0168-8227. doi: 10.1016/j.diabres.2024.111886. URL http://dx.doi.org/10.1016/j.diabres.2024.111886.

Rachel G. Curtis, Dorothea Dumuid, Timothy Olds, Ronald Plotnikoff, Corneel Vandelanotte, Jillian Ryan, Sarah Edney, and Carol Maher. The association between time-use behaviors and physical and mental well-being in adults: A compositional isotemporal substitution analysis. *Journal of Physical Activity and Health*, 17(2):197–203, February 2020. ISSN 1543-5474. doi: 10.1123/jpah.2018-0687. URL http://dx.doi.org/10.1123/jpah.2018-0687.

Dorothea Dumuid, Željko Pedišić, Tyman Everleigh Stanford, Josep-Antoni Martín-Fernández, Karel Hron, Carol A Maher, Lucy K Lewis, and Timothy Olds. The compositional isotemporal substitution model: A method for estimating changes in a health outcome for reallocation of time between sleep, physical activity and sedentary behaviour. *Statistical Methods in Medical Research*, 28(3):846–857, November 2017. ISSN 1477-0334. doi: 10.1177/0962280217737805. URL http://dx.doi.org/10.1177/0962280217737805.

Stuart J. Fairclough, Dorothea Dumuid, Sarah Taylor, Whitney Curry, Bronagh McGrane, Gareth Stratton, Carol Maher, and Timothy Olds. Fitness, fatness and the reallocation of time between children's daily movement behaviours: an analysis of compositional data. *International Journal of Behavioral Nutrition and Physical Activity*, 14(1), May 2017. ISSN 1479-5868. doi: 10.1186/s12966-017-0521-z. URL http://dx.doi.org/10.1186/s12966-017-0521-z.

Aina M. Galmes-Panades, Veronica Varela-Mato, Jadwiga Konieczna, Julia Wärnberg, Miguel Ángel Martínez-González, Jordi Salas-Salvadó, Dolores Corella, Helmut Schröder, Jesús Vioque, Ángel M. Alonso-Gómez, J. Alfredo Martínez, Luís Serra-Majem, Ramon Estruch, Francisco J. Tinahones, José Lapetra, Xavier Pintó, Josep A. Tur, Antonio Garcia-Rios, Blanca Riquelme-Gallego, José Juan Gaforio, Pilar Matía-Martín, Lidia Daimiel, Rafael Manuel Micó Pérez, Josep Vidal, Clotilde Vázquez, Emilio Ros, Ana Garcia-Arellano, Andrés Díaz-López, Eva M. Asensio, Olga Castañer, Francisca Fiol, Luis Alfredo Mira-Castejón, Anai Moreno Rodríguez, Juan Carlos Benavente-Marín, Itziar Abete, Laura Tomaino, Rosa Casas, F. Javier Barón López, José Carlos Fernández-García, José Manuel Santos-Lozano, Ana Galera, Catalina M. Mascaró, Cristina Razquin, Christopher Papandreou, Olga Portoles, Karla Alejandra Pérez-Vega, Miguel Fiol, Laura Compañ-Gabucio, Jessica Vaquero-Luna, Miguel Ruiz-Canela, Nerea Becerra-Tomás, Montserrat Fitó, and Dora Romaguera. Isotemporal substitution of inactive time with physical activity and time in bed: cross-sectional associations with cardiometabolic health in the predimed-plus study. *International Journal of Behavioral Nutrition and Physical Activity*, 16(1), December 2019. ISSN 1479-5868. doi: 10.1186/s12966-019-0892-4. URL http://dx.doi.org/10.1186/s12966-019-0892-4.

Aina M Galmes-Panades, Jadwiga Konieczna, Veronica Varela-Mato, Itziar Abete, Nancy Babio, Miquel Fiol, Jose Antonio de Paz, Rosa Casas, Romina Olbeyra, Miguel Ruiz-Canela, Antoni Palau Galindo, Olga Castañer, Arturo Martín-García, Ramon Estruch, Josep Vidal, Pilar Buil-Cosiales, Julia Wärnberg, Jordi Salas-Salvado, J Alfredo Martinez, and Dora Romaguera. Changes in physical activity, sedentary behaviour and body composition: longitudinal analysis in the predimed-plus trial. July 2020. doi: 10.21203/rs.3.rs-44115/v1. URL http://dx.doi.org/10.21203/rs.3.rs-44115/v1.

Aina M. Galmes-Panades, Jadwiga Konieczna, Veronica Varela-Mato, Itziar Abete, Nancy Babio, Miquel Fiol, José Antonio de Paz, Rosa Casas, Romina Olbeyra, Miguel Ruiz-Canela, Antoni Palau-Galindo, Olga Castañer, Arturo Martín-García, Ramón Estruch, Josep Vidal, Pilar Buil-Cosiales, Julia Wärnberg, Jordi Salas-Salvadó, J. Alfredo Martínez, Dora Romaguera, V. Martin, R. Pedret-Llaberia, R. Gonzalez, R. Sagarra-Álamo, F. París-Palleja, J. Balsells, J. M. Roca, T. Basora-Gallisa, J. Vizcaino, P. Llobet-Alpizarte, C. Anguera-Perpiñá, M. Llauradó-Vernet, C. Caballero, M. Garcia-Barco, M. D. Morán-Martínez, J. García-Rosselló, A. del Pozo, C. Poblet-Calaf, P. Arcelin-Zabal, X. Floresví, M. Ciutat-Benet, J. J. Cabré-Vila, F. Dolz-Andrés, M. Soler, M. Garcia-Vidal, J. Vilalta, J. Boj Casajuana, M. Ricard, F. Saiz, A. Isach, M. Sanchez-Marin Martinez, E. Granado-Font, C. Lucena-Luque, C. Mestres-Sola, N. Becerra-Tomás, G. Mestres, J. Basora, G. Mena-Sánchez, L. Barrubés-Piñol, M. Gil-Segura, N. Rosique-Esteban, S. Chig, I. Abellán-Cano, V. Ruiz-García, C. Gomez-Martinez, L. Lopez-Gonzalez, A. Salas-Huetos, I. Paz-Graniel, J. Roig-Vallverdú, C. Miñana-Garcia, L. Sánchez-Niembro, P. Hernandez-Alonso, S. Canudas, A. Díaz-López, E. Toledo, M. A. Martínez-González, Z. Vázquez, C. Razquin, M. Bes-Rastrollo, A. Gea, A. Sanchez-Tainta, B. Sanjulián-Aranguren, E. Goñi, L. Goñi, M. J. Cobo, A. Rico-Campa, F. J. Basterra-Gortari, A. Garcia-Arellano, J. Diez-Espino, O. Lecea-Juarez, J. Carlos Cenoz-Osinaga, I. Alvarez-Alvarez, M. C. Sayon-Orea, C. I. Fernandez-Lázaro, L. Ruiz-Estigarribia, J. Bartolome-Resano, A. Sola-Larraza, E. Lozano-Oloriz, B. Cano-Valles, S. Eguaras, E. Pascual Roquet-Jalmar, I. Galilea-Zabalza, H. Lancova, R. Ramallal, M. L. Garcia-Perez, V. Estremera-Urabayen, M. J. Ariz-Arnedo, C. Hijos-Larraz, C. Fernandez-Alfaro, B. Iñigo-Martinez, R. Villanueva-Moreno, S. Martin-Almendros, L. Barandiaran-Bengoetxea, C. Fuertes-Goñi, A. Lezaun-Indurain, M. J. Guruchaga-Arcelus, O. Olmedo-Cruz, L. Escriche-Erviti, R. Ansorena-Ros, R. Sanmatin-Zabaleta, J. Apalategi-Lasa, J. Villanueva-Telleria, M. M. Hernández-Espinosa, L. Herrera-Valdez, L. Dorronsoro-Dorronsoro, L. Echeverria-Lizarraga, J. A. Cabeza-Beunza, P. Fernández-Urretavizcaya, P. Gascó-García, C. Royo-Jimenez, J. Moran-Pí, F. Salazar-Fernández, F. J. Chasco-Ros, F. Cortés-Ugalde, J. J. Jurio-Burgui, P. Pascual-Pascual, A. I. Rodríguez-Ezpeleta, M. Esparza-Cáceres, C. Arroyo-Azpa, M. Rodríguez-Sanz de Galdeano, T. Forcen-Alonso, M. Armendariz-Marcotegui, A. Brugos-Larumbe, A. Arillo, B. López-Aisa, M. Moñino, A. Colom, M. Morey, M. A. Martín, E. Rayó, J. Llobera, C. Fernández-Palomeque, E. Fortuny, M. Noris, L. López, X. Rosselló, S. Munuera, F. Tomás, F. Fiol, A. Jover, J. M. Janer, C. Vallespir, I. Mattei, N. Feuerbach, M. M. Sureda, S. Vega, L. Quintana, A. Fiol, M. Amador, S. González, J. Coll, A. Moyá, T. Piqué-Sistac, M. D. Sanmartín-Fernández, M. C. Piña-Valls, M. A. Llorente San Martín, J. Pou-Bordoy, I. Cantero, C. Cristobo, I. Ibero-Baraibar, M. Zulet, J. Ágreda Peiró, M. D. Lezáun-Burgui, N. Goñi-Ruiz, R. Bartolomé-Resano, E. Cano-Cáceres, T. Elcarte-López, E. Echarte-Osacain, B. Pérez-Sanz, I. Blanco-Platero, A. Andueza-Azcárate, A. Gimeno-Aznar, E. Ursúa-Sesma, B. Ojeda-Bilbao, J. Martinez-Jarauta, L. Ugalde-Sarasa, B. Rípodas-Echarte, M. V. Güeto-Rubio, C. Napal-Lecumberri, M. D. Martínez-Mazo, E. Arina-Vergara, A. Parra-Osés, F. Artal-Moneva, F. Bárcena-Amigo, F. Calle-Irastoza, J. Abad-Vicente, J. I. Armendáriz-Artola, P. Iñigo-Cibrian, J. Escribano-Jarauta, J. Ulibarri-delportillo, B. Churio-Beraza, Y. Monzón-Martínez, E. Madoz-Zubillaga, C. Arroniz, C. Viñas, S. Castro-Barquero, A. M. Ruiz-León, R. Losno, L. Tarés, A. Jordán, R. Soriano, M. Camafort, C. Sierra, E. Sacanella, J. M. Cots, I. Sarroca, M. García, N. Bermúdez, A. Pérez, I. Duaso, A. de la Arada, R. Hernández, C. Simón, M. A. de la Poza, I. Gil, M. Vila, C. Iglesias, N. Assens, M. Amatller, L. L. Rams, T. Benet, G. Fernández, J. Teruel, A. Azorin, M. Cubells, D. López, J. M. Llovet, M. L. Gómez, P. Climente, L. de Paula, J. Soto, C. Carbonell, C. Llor, X. Abat, A. Cama, M. Fortuny, C. Domingo, A. I. Liberal, T. Martínez, E. Yañez, M. J. Nieto, A. Pérez, E. Lloret, C. Carrazoni, A. M. Belles, C. Olmos, M. Ramentol, M. J. Capell, R. Casas, I. Giner,

A. Muñoz, R. Martín, E. Moron, A. Bonillo, G. Sánchez, C. Calbó, J. Pous, M. Massip, Y. García, M. C. Massagué, R. Ibañez, J. Llaona, T. Vidal, N. Vizcay, E. Segura, C. Galindo, M. Moreno, M. Caubet, J. Altirriba, G. Fluxà, P. Toribio, E. Torrent, J. J. Anton, A. Viaplana, G. Vieytes, N. Duch, A. Pereira, M. A. Moreno, E. Sant, J. Gené, H. Calvillo, F. Pont, M. Puig, M. Casasayas, A. Garrich, E. Senar, A. Martínez, I. Boix, E. Sequeira, V. Aragunde, S. Riera, M. Salgado, M. Fuentes, E. Martín, A. Ubieto, F. Pallarés, C. Sala, A. Abilla, S. Moreno, E. Mayor, T. Colom, A. Gaspar, A. Gómez, L. Palacios, R. Garrigosa, V. Martín, S. Abajo-Olea, L. Álvarez Álvarez, M. Rubín-García, A. Torres, P. Farias, N. Cubelos, A. Adlbi Sibai, M. Ajenjo, E. Carriedo-Ule, M. Escobar-Fernández, J. I. Ferradal-García, J. P. Fernández-Vázquez, C. González-Quintana, F. González-Rivero, M. Lavinia-Popescu, J. I. López-Gil, J. López de la Iglesia, A. Marcos-Delgado, C. Merino-Acevedo, S. Reguero-Celada, M. Rodríguez-Bul, E. Fernández-Mielgo, A. Altés, I. Vinagre, C. Mestre, J. Viaplana, M. Serra, J. Vera, T. Freitas, E. Ortega, and I. Pla. Targeting body composition in an older population: do changes in movement behaviours matter? longitudinal analyses in the predimed-plus trial. *BMC Medicine*, 19(1), January 2021. ISSN 1741-7015. doi: 10.1186/s12916-020-01847-9. URL `http://dx.doi.org/10.1186/s12916-020-01847-9`.

Jozo Grgic, Dorothea Dumuid, Enrique Garcia Bengoechea, Nipun Shrestha, Adrian Bauman, Timothy Olds, and Zeljko Pedisic. Health outcomes associated with reallocations of time between sleep, sedentary behaviour, and physical activity: a systematic scoping review of isotemporal substitution studies. *International Journal of Behavioral Nutrition and Physical Activity*, 15(1), July 2018. ISSN 1479-5868. doi: 10.1186/s12966-018-0691-3. URL `http://dx.doi.org/10.1186/s12966-018-0691-3`.

Shawn Hakimi, Luc J. Martin, and Mark W. Rosenberg. Impacts of physical activity, sedentary behaviour, and sleep on depression symptoms in canadian older adults 65 years of age and above: a compositional data analysis of the canadian longitudinal study on aging. *Journal of Activity, Sedentary and Sleep Behaviors*, 3(1), March 2024. ISSN 2731-4391. doi: 10.1186/s44167-024-00047-7. URL `http://dx.doi.org/10.1186/s44167-024-00047-7`.

Amy Hofman, Trudy Voortman, M. Arfan Ikram, and Annemarie I Luik. Substitutions of physical activity, sedentary behaviour and sleep: associations with mental health in middle-aged and elderly persons. *Journal of Epidemiology and Community Health*, 76(2):175–181, July 2021. ISSN 1470-2738. doi: 10.1136/jech-2020-215883. URL `http://dx.doi.org/10.1136/jech-2020-215883`.

Ian Janssen, Anna E. Clarke, Valerie Carson, Jean-Philippe Chaput, Lora M. Giangregorio, Michelle E. Kho, Veronica J. Poitras, Robert Ross, Travis J. Saunders, Amanda Ross-White, and Sebastien F.M. Chastin. A systematic review of compositional data analysis studies examining associations between sleep, sedentary behaviour, and physical activity with health outcomes in adults. *Applied Physiology, Nutrition, and Metabolism*, 45(10 (Suppl. 2)):S248–S257, October 2020. ISSN 1715-5320. doi: 10.1139/apnm-2020-0160. URL `http://dx.doi.org/10.1139/apnm-2020-0160`.

Luís Lemos, Cain Clark, Caroline Brand, Maria Luíza Pessoa, Anelise Gaya, Jorge Mota, Michael Duncan, and Clarice Martins. 24-hour movement behaviors and fitness in preschoolers: A compositional and isotemporal reallocation analysis. *Scandinavian Journal of Medicine amp; Science in Sports*, 31(6):1371–1379, March 2021. ISSN 1600-0838. doi: 10.1111/sms.13938. URL `http://dx.doi.org/10.1111/sms.13938`.

Zhaoxu Lu, Jin Guo, Chuanhe Liu, Jian Wu, Chuo Zhao, Fang Wang, Yihua Bao, Haowen Zhang, Bing Qi, Xiuhe Li, Hongyan Guan, Jianxin Wu, and Ting Zhang. Reallocation of time to moderate-to-vigorous physical activity and estimated changes in physical fitness among preschoolers: a compositional data analysis. *BMC Public Health*, 24(1), October 2024. ISSN 1471-2458. doi: 10.1186/s12889-024-20290-6. URL http://dx.doi.org/10.1186/s12889-024-20290-6.

Jaclyn P. Maher, Alexander Jochim, Yeongjun Seo, Brynn L. Hudgins, Iman F. Khan, Peyton A. Greco, Emily M. Postlethwait, Om L. Patel, Krista S. Leonard, Chih-Hsiang Yang, and Denver M. Y. Brown. Longitudinal associations between waking movement behaviors and health-related quality of life in older adults: a compositional data analysis approach. *Journal of Activity, Sedentary and Sleep Behaviors*, 4(1), July 2025. ISSN 2731-4391. doi: 10.1186/s44167-025-00080-0. URL http://dx.doi.org/10.1186/s44167-025-00080-0.

Duncan E. McGregor, Valerie Carson, Javier Palarea-Albaladejo, Philippa M. Dall, Mark S. Tremblay, and Sebastien F. M. Chastin. Compositional analysis of the associations between 24-h movement behaviours and health indicators among adults and older adults from the canadian health measure survey. *International Journal of Environmental Research and Public Health*, 15(8):1779, August 2018. ISSN 1660-4601. doi: 10.3390/ijerph15081779. URL http://dx.doi.org/10.3390/ijerph15081779.

Luiz Menezes-Júnior. Reallocating wakeful movement behaviors and sleep health: A partial wakeful day substitution model analysis of 27,048 brazilian adults. *Sleep Medicine*, 146:109051, October 2026. ISSN 1389-9457. doi: 10.1016/j.sleep.2026.109051. URL http://dx.doi.org/10.1016/j.sleep.2026.109051.

Jairo H. Migueles, Christine Delisle Nyström, Marja H. Leppänen, Pontus Henriksson, and Marie Löf. Revisiting the cross-sectional and prospective association of physical activity with body composition and physical fitness in preschoolers: A compositional data approach. *Pediatric Obesity*, 17(8), February 2022. ISSN 2047-6310. doi: 10.1111/ijpo.12909. URL http://dx.doi.org/10.1111/ijpo.12909.

Gunchmaa Nyam, Natalie Lander, Ana Maria Contardo Ayala, Claudia Strugnell, and Jo Salmon. Associations between physical activity and sedentary time with sleep duration and sleep quality in adolescents. April 2026. doi: 10.21203/rs.3.rs-9205166/v1. URL http://dx.doi.org/10.21203/rs.3.rs-9205166/v1.

Rosario Ortolá, Esther García-Esquinas, Verónica Cabanas-Sánchez, Jairo H Migueles, David Martínez-Gómez, and Fernando Rodríguez-Artalejo. Association of physical activity, sedentary behavior, and sleep with unhealthy aging: Consistent results for device-measured and self-reported behaviors using isotemporal substitution models. *The Journals of Gerontology: Series A*, 76(1):85–94, July 2020. ISSN 1758-535X. doi: 10.1093/gerona/glaa177. URL http://dx.doi.org/10.1093/gerona/glaa177.

Jana Pelclová, Jan Vindiš, Paulína Jašková, and Karel Hron. The interplay between lying, sitting, standing, moving, and walking on obesity risk in older adults: a compositional and isotemporal substitution analysis. *BMC Geriatrics*, 24(1), December 2024. ISSN 1471-2318. doi: 10.1186/s12877-024-05619-5. URL http://dx.doi.org/10.1186/s12877-024-05619-5.

Huiqi Song, Borui Zhang, Nike Lu, Yang Liu, and Patrick W. C. Lau. Longitudinal associations between 24-hour movement behaviors and physical fitness in preschoolers: a compositional

isotemporal substitution analysis. *Scientific Reports*, 15(1), September 2025. ISSN 2045-2322. doi: 10.1038/s41598-025-01231-8. URL http://dx.doi.org/10.1038/s41598-025-01231-8.

Christine W. St. Laurent, Sarah Burkart, Katrina Rodheim, Robert Marcotte, and Rebecca M. C. Spencer. Cross-sectional associations of 24-hour sedentary time, physical activity, and sleep duration compositions with sleep quality and habits in preschoolers. *International Journal of Environmental Research and Public Health*, 17(19):7148, September 2020. ISSN 1660-4601. doi: 10.3390/ijerph17197148. URL http://dx.doi.org/10.3390/ijerph17197148.

Emmanuel Stamatakis, Kris Rogers, Ding Ding, David Berrigan, Josephine Chau, Mark Hamer, and Adrian Bauman. All-cause mortality effects of replacing sedentary time with physical activity and sleeping using an isotemporal substitution model: a prospective study of 201,129 mid-aged and older adults. *International Journal of Behavioral Nutrition and Physical Activity*, 12(1), September 2015. ISSN 1479-5868. doi: 10.1186/s12966-015-0280-7. URL http://dx.doi.org/10.1186/s12966-015-0280-7.

Emmanuel Stamatakis, Joanne Gale, Adrian Bauman, Ulf Ekelund, Mark Hamer, and Ding Ding. Sitting time, physical activity, and risk of mortality in adults. *Journal of the American College of Cardiology*, 73(16):2062–2072, April 2019. ISSN 0735-1097. doi: 10.1016/j.jacc.2019.02.031. URL http://dx.doi.org/10.1016/j.jacc.2019.02.031.

Haoyang Sun, Zhaoxu Lu, Jin Guo, Yihua Bao, Haowen Zhang, Fang Wang, Xiangli Liu, Jianwei Li, Yanpeng Ni, Di Heng, Haiyuan Zhou, Jian Wu, and Ting Zhang. Compositional isotemporal substitution analysis of 24-h movement behaviours and speed capability in preschoolers. *Child: Care, Health and Development*, 52(1), December 2025. ISSN 1365-2214. doi: 10.1111/cch.70218. URL http://dx.doi.org/10.1111/cch.70218.

Deborah M. Telford, Rebecca M. Meiring, and Silmara Gusso. Sitting less, moving more: A compositional data analysis of movement behaviours and adolescent health-related quality of life. *Acta Paediatrica*, 115(6):1266–1273, February 2026. ISSN 1651-2227. doi: 10.1111/apa.70488. URL http://dx.doi.org/10.1111/apa.70488.

Mark A. Tully, Ilona McMullan, Nicole E. Blackburn, Jason J. Wilson, Brendan Bunting, Lee Smith, Frank Kee, Manuela Deidda, Maria Giné-Garriga, Laura Coll-Planas, Dhayana Dallmeier, Michael Denkinger, Dietrich Rothenbacher, and Paolo Caserotti. Sedentary behavior, physical activity, and mental health in older adults: An isotemporal substitution model. *Scandinavian Journal of Medicine amp; Science in Sports*, 30(10):1957–1965, July 2020. ISSN 1600-0838. doi: 10.1111/sms.13762. URL http://dx.doi.org/10.1111/sms.13762.

Julie Vanderlinden, Gregory Biddle, Filip Boen, and Jannique van Uffelen. Are reallocations between sedentary behaviour and physical activity associated with better sleep in adults aged 55+ years? an isotemporal substitution analysis. *International Journal of Environmental Research and Public Health*, 17(24):9579, December 2020. ISSN 1660-4601. doi: 10.3390/ijerph17249579. URL http://dx.doi.org/10.3390/ijerph17249579.

Shuai Wang, Wei Liang, Huiqi Song, Ning Su, Lin Zhou, Yanping Duan, Ryan E. Rhodes, Huaxuan Liu, Yi-de Yang, Wing Chung Patrick Lau, and Julien Steven Baker. Prospective association between 24-hour movement behaviors and mental health among overweight/obese college students: a compositional data analysis approach. *Frontiers in Public Health*, 11, October 2023. ISSN 2296-2565. doi: 10.3389/fpubh.2023.1203840. URL http://dx.doi.org/10.3389/fpubh.2023.1203840.

Jyh Eiin Wong, Javier Palarea-Albaladejo, Shoo Thien Lee, Denise Koh, Ilse Khouw, Bee Koon Poh, and . *Association between 24−hour movement behaviors and adiposity in malaysian school children: A compositional isotemporal substitution analys* URL http://dx.doi.org/10.1123/jpah.2024-0161.

Wenya Zhang, Yang Pan, Yiwen Dai, Jie Liang, Jingya Ma, Yuling Liu, Darui Gao, Yanyu Zhang, Mengmeng Ji, Wuxiang Xie, and Fanfan Zheng. Reallocation of time between accelerometer-derived movement behaviors, genetic susceptibility, and risk of incident dementia, mortality, and premature death: a longitudinal cohort study. *International Journal of Behavioral Nutrition and Physical Activity*, 22(1), August 2025. ISSN 1479-5868. 10.1186/s12966-025-01814-8. URL http://dx.doi.org/10.1186/s12966-025-01814-8.

Yuxuan Zhang, Jiaqiao Zhang, Jiaxin Yue, Fan Liu, Xiuxiu Huang, and Dandan Jia. Reallocating 24-hour movement behaviors and cardiometabolic health in adults: a systematic review and meta-analysis. *Archives of Public Health*, 84(1), March 2026. ISSN 2049-3258. 10.1186/s13690-026-01885-4. URL http://dx.doi.org/10.1186/s13690-026-01885-4.